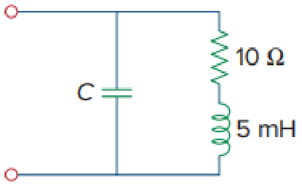
Chapter 9, Problem 89CP

Problem

An industrial load is modeled as a series combination of an inductor and a resistance as shown in Fig. 9.89. Calculate the value of a capacitor C across the series combination so that the net impedance is resistive at a frequency of 2 kHz.

Figure 9.89:



Step-by-step solution

Step 1 of 4

Draw the circuit diagram as shown below.

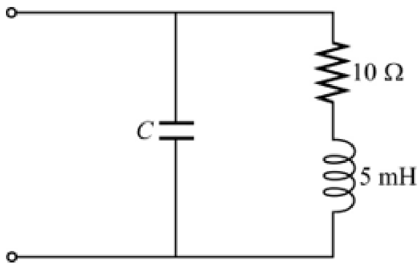


Figure 1

Step 2 of 4

Calculate the angular frequency.

$$\begin{aligned}\omega &= 2\pi f \\ &= (2)(3.14)\left(2\times10^{3}\right) \\ &= 12.566\times10^{3}\ \text{rad/s}\end{aligned}$$

Resistance $R = 10\ \Omega$ and inductance $L = 5\ \text{mH}$.

Inductor, $5\ \text{mH}$ is in series with the resistor $10\ \Omega$. The impedance combination is in parallel with the capacitor.

Given net impedance is resistive at a frequency of $2\ \text{kHz}$.

$$\begin{aligned}Z_{eq} \text{ or } Z_{\text{net impedance}} &= \frac{(R+j\omega L)\left(\dfrac{1}{j\omega C}\right)}{(R+j\omega L)+\dfrac{1}{j\omega C}} \\ &= \frac{R+j\omega L}{j\omega RC + j^2\omega^2 LC + 1} \\ &= \frac{R+j\omega L}{\left(1-\omega^2 LC\right)+j\omega RC}\times\frac{\left(1-\omega^2 LC\right)-j\omega RC}{\left(1-\omega^2 LC\right)-j\omega RC} \\ &= \frac{R\left(1-\omega^2 LC\right)-j\omega R^2 C + j\omega L\left(1-\omega^2 LC\right)+\omega^2 LRC}{\left(1-\omega^2 LC\right)^2+\left(\omega RC\right)^2}\end{aligned}$$

Step 3 of 4

Simplify further.

$$Z_{eq} \text{ or } Z_{\text{net impedance}} = \frac{R\left(1-\omega^2 LC\right)+\omega^2 LRC + j\left(\omega L\left(1-\omega^2 LC\right)-\omega R^2 C\right)}{\left(1-\omega^2 LC\right)^2+\left(\omega RC\right)^2}$$

The net impedance is resistive that means imaginary part is zero.

$$\begin{aligned}&\frac{\omega L\left(1-\omega^2 LC\right)-\omega R^2 C}{\left(1-\omega^2 LC\right)^2+\left(\omega RC\right)^2} = 0 \\ &\omega L\left(1-\omega^2 LC\right)-\omega R^2 C = 0 \\ &\omega L - \omega^3 L^2 C = \omega R^2 C \\ &L - \omega^2 L^2 C = R^2 C \\ &C\left(R^2+\omega^2 L^2\right) = L\end{aligned}$$

Step 4 of 4

Simplify further.

$$C = \frac{L}{R^2+\omega^2 L^2}$$

Substitute 10 for R, 5×10^{3} for L and 12.566×10^{3} for ω in the equation.

$$\begin{aligned}C &= \frac{5\times10^{-3}}{100+\left(2\pi\times2\times10^{3}\right)^2\left(5\times10^{-3}\right)^2} \\ &= 1.233\times10^{-6}\ \text{F} \\ &= 1.233\ \mu\text{F}\end{aligned}$$

Therefore, the value of capacitance C is $\boxed{1.233\ \mu\text{F}}$.